

Title

Early Wildfire Risk Prediction Using Near Real-Time NASA FIRMS Satellite Event Data

Abstract

Early detection and timely assessment of wildfire risk are critical for minimizing environmental damage and enabling rapid emergency response. Most existing wildfire monitoring systems rely on satellite imagery, video surveillance, or deep learning–based image analysis, which are computationally intensive and often detect fires only after ignition or significant damage has occurred. This paper proposes a novel **event-based early wildfire risk assessment framework** using **NASA FIRMS Near Real-Time (NRT) fire event data**. Instead of image processing, the proposed approach analyzes spatiotemporal fire event streams using key attributes such as brightness temperature, fire radiative power (FRP), detection confidence, spatial proximity, and temporal persistence. A new **Fire Alert Readiness Index (FARI)** is introduced to classify fire events into actionable risk levels, enabling early identification of potential wildfire escalation while reducing false alarms. The framework emphasizes computational efficiency, scalability, and interpretability, making it suitable for real-time operational deployment. By shifting from image-based detection to event-level spatiotemporal modeling, the proposed system fills a critical research gap and provides an effective decision-support tool for early wildfire monitoring and risk management.